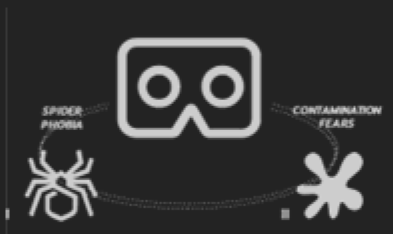
Comparative Effectiveness of Virtual Reality (VR) and Augmented Reality (AR) in Virtual Exposure Therapy



BACKGROUND

Exposure Therapy

Evidence-based treatment for specific phobias: exposure therapy for a patient with spider phobia. (© Institute for Translational Psychiatry, University of Münster)

Limitations

- difficult to set up
- safety risks
- mental pressure

Ali in the Mind presenter Sana Qadar undergoes a simulated VR therapy session (ABC Radio National: James Bullen)

Previous research - Supported the efficacy - Either VR or AR independently - Choose which technology to use?

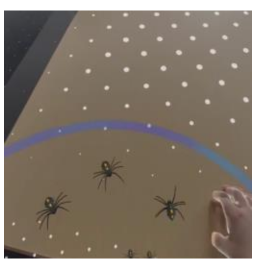
RESEARCH QUESTIONS

Q1. Is there a significant difference in the effectiveness of AR and VR when used in exposure therapy?

Q2. If a difference exists, is one generally better than the other, or do the results vary depending on the specific task?

STUDY DESIGN

Within-subjects



Project 1
Spider Phobia Exposure

AR | VR

Project 2
Contamination Fear Exposure

AR | VR

RESULTS

Hypothesis Testing

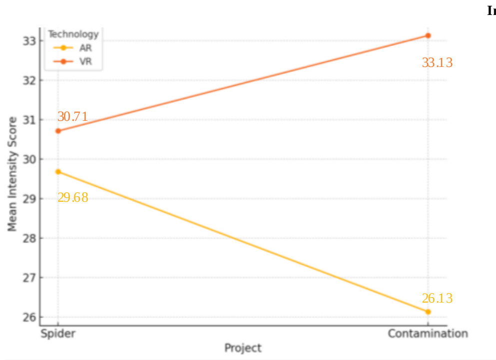
	Conditions	*Mean*	*SD*
N = 31	***Spider _ AR***	29.68	9.758
	Spider _ VR	30.71	10.143
	Contamination _ AR	26.13	10.941
	Contamination _ VR	33.13	11.215

Main Effect of Technology:

$F_{(1, 30)} = 14.358, p < 0.001, \eta^2 = 0.324$

There is a significant difference in the intensity of experiences provided by AR and VR technologies

Q1 - VR generally offering a slightly higher intensity of experience than AR.



Interaction Effect (Technology * Project):

$F_{(1, 30)} = 10.251, p = 0.003, \eta^2 = 0.255$

The impact of technology on the intensity of experiences differs significantly between the two projects

Post-hoc test

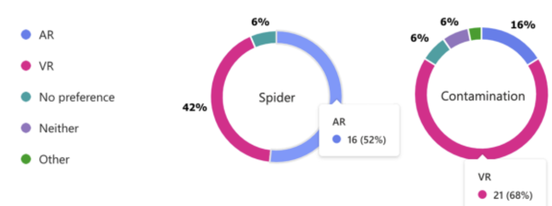
Spider Phobia: $t(11) = -0.73$,
$p = 0.471$ (no significant difference);
Contamination Fear: $t(11) = -4.97$,
$p < 0.001$ (significant difference)

Q2 - The performance of different technologies varied between projects.

Although VR had higher average scores in both projects, it was more effective in contamination. AR provided a relatively stronger experience in spider.

Technology Preference

Regarding exposure therapy for spider phobia / contamination fears, based on your experience, do you prefer VR or AR technology?



Participant Feedback - Contamination Project

- **Environmental Atmosphere** — *When the source of fear not only comes from the object itself but is also influenced by the surrounding atmosphere, the VR environment helps create a sense of fear and unease, triggering a stronger emotional response. And AR provides users with a certain level of control and detachment, allowing them to manage their fear by avoiding interaction. In contrast, VR fully immerses users, surrounding them in a passive, immersive environment.*
- **Multi-Sensory Illusions** — *In VR, visuals are so vivid that they trigger psychological responses in other senses, such as smell and touch, even without direct sensory stimuli. The brain fills in the gaps, strengthening emotional reactions.*
- **Environmental Coherence** — *In VR, the environment is fully controlled and designed to create a coherent and consistent scene. In a dirty setting, all elements like rubbishes are part of an immersive world, making it easier to psychologically accept the realism of the scene. In AR, however, there's a disconnect between the real physical world and the virtual elements.*
- **Memory Triggering** — *In VR, realistic and immersive environments are more likely to trigger personal memories related to similar real-life experiences. By recreating familiar or memory-linked settings, VR has the potential to tap into deeper emotional responses.*

- Spider Project

- **Blur and Dynamic Realism** — *In real life, people rarely focus on spiders closely; instead, they often sense them peripherally or with anxious glances. The blurred motion in AR mirrors this natural fear response, enhancing the realism and emotional impact of the experience. Also, AR uses real-world lighting conditions and physical effects to 're-render' virtual objects, causing objects (like the virtual spider) to become blurry when moving. Participants pointed out that this blurriness made the spider appear less like an artificial creation and more like a real creature.*
- **Passive Threat Perception** — *In AR, participants continuously perceive a threat from the virtual spider, even when it is static, as they can see both the spider and their own body in the real world. This passive threat is effective because the virtual spider, like a real creature, creates an expectation of potential movement. In VR, the fully digital environment, with its lower sense of embodiment, may reduce this perceived threat.*
- **Enhanced Sense of Bodily Presence** — *AR allows participants to see and feel their real body interacting with virtual elements, making the experience more lifelike. As P19 pointed out, seeing their own hands in AR made it easier to believe they were actually holding a spider, not just interacting with a virtual object.*
- **Influence of VR Gaming Experience** — *The widespread adoption of VR gaming influenced participants' responses to potential fear-inducing elements during exposure therapy. Previous gaming experience can lead to a 'gamer mindset', reducing the likelihood of fear or anxiety reactions.*

DISCUSSION

AR | **VR**

There is a significant difference between AR and VR, with results varying based on the specific task.

Development Suggestion: Gradual and Progressive Therapy Combining AR and VR

Adaptation Period
Use of AR in the Initial Stage

Intensification Period
VR as the Treatment Progresses

Dynamic Period
Adjust and Deepen Exposure Treatment

[**AR** + **VR**]